Quantum Photonics Integrated Circuit Specification

Document Reference: QPIC-TR-2026-V4 | Page 1 Architecture Overview

Photonic quantum processors harness discrete photons as information-carrying qubits, leveraging integrated silicon nitride (SiN) waveguides for phase-stable spatial routing. The optical transmission loss of the primary waveguide bus is precisely calibrated to 0.12 dB/cm across the telecommunication C-band spectrum. A reconfigurable triangular lattice of Mach-Zehnder interferometers provides universal linear optical unitaries.

Phase modulation is executed through thermo-optic phase shifters with a switching rise time of 1.4 microseconds and low cross-talk below -34 dB. Cryogenic coupling is achieved through inverted taper edge couplers boasting an edge-coupling insertion loss of 0.8 dB per facet.

Detection Subsystem & Cryostat Calibration

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Detection occurs using Superconducting Nanowire Single-Photon Detectors (SNSPDs) micro-fabricated from amorphous molybdenum silicide (MoSi) thin films. The detection system achieves a peak quantum efficiency of 98.4% at 1550 nm telecommunication wavelength, operating in a closed-cycle pulse-tube cryostat maintaining a stable base temperature of 0.85 Kelvin. Timing jitter is suppressed below 14.8 picoseconds FWHM.

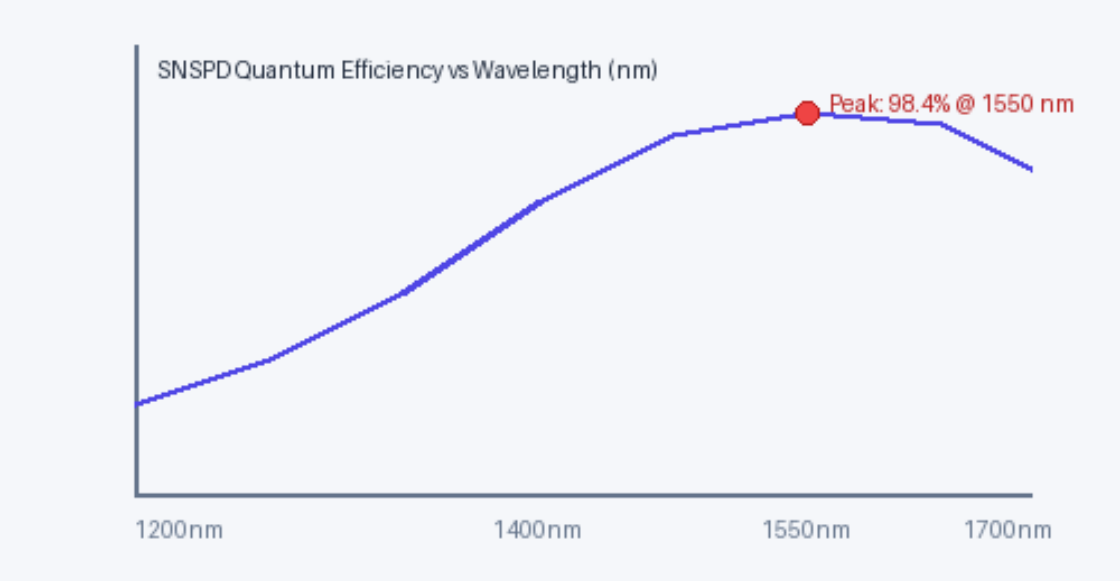

Figure 2.1: Calibrated Single-Photon Detector Quantum Efficiency across the 1200-1700 nm optical band. Notice the high-fidelity resonant plateau centered at 1550 nm.